```julia
using DataFrames
```

```julia
using CSV
```

```julia
using Plots
```

```julia
using Statistics
```

```julia
using StatsPlots
```

```
url = "https://people.sc.fsu.edu/~jburkardt/data/csv/airtravel.csv"
```

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url = "https://people.sc.fsu.edu/~jburkardt/data/csv/airtravel.csv"
```

df =

	Month	1958	1959	1960
1	"JAN"	340	360	417
2	"FEB"	318	342	391
3	"MAR"	362	406	419
4	"APR"	348	396	461
5	"MAY"	363	420	472
6	"JUN"	435	472	535
7	"JUL"	491	548	622
8	"AUG"	505	559	606
9	"SEP"	404	463	508
10	"OCT"	359	407	461
11	"NOV"	310	362	390
12	"DEC"	337	405	432

```julia
df = CSV.read(download(url), DataFrame)
```

```
passengers_data = 12×3 Matrix{Int64}:
 340  360  417
 318  342  391
 362  406  419
 348  396  461
 363  420  472
 435  472  535
 491  548  622
 505  559  606
 404  463  508
 359  407  461
 310  362  390
 337  405  432
```

```
passengers_data = Matrix(df[:, 2:end])
```

```
mean_passengers = 12×1 Matrix{Float64}:
                  372.3333333333333
                  350.3333333333333
                  395.6666666666667
                  401.6666666666667
                  418.3333333333333
                  480.6666666666667
                  553.6666666666666
                  556.6666666666666
                  458.3333333333333
                  409.0
                  354.0
                  391.3333333333333
```

```
mean_passengers = mean(passengers_data, dims=2)
```

```
[372.333, 350.333, 395.667, 401.667, 418.333, 480.667, 553.667, 556.667, 458.333, 409.0, 3
```

```
df.MeanPassengers_vec = vec(mean_passengers)
```

```
[372.333, 350.333, 395.667, 401.667, 418.333, 480.667, 553.667, 556.667, 458.333, 409.0, 3
```

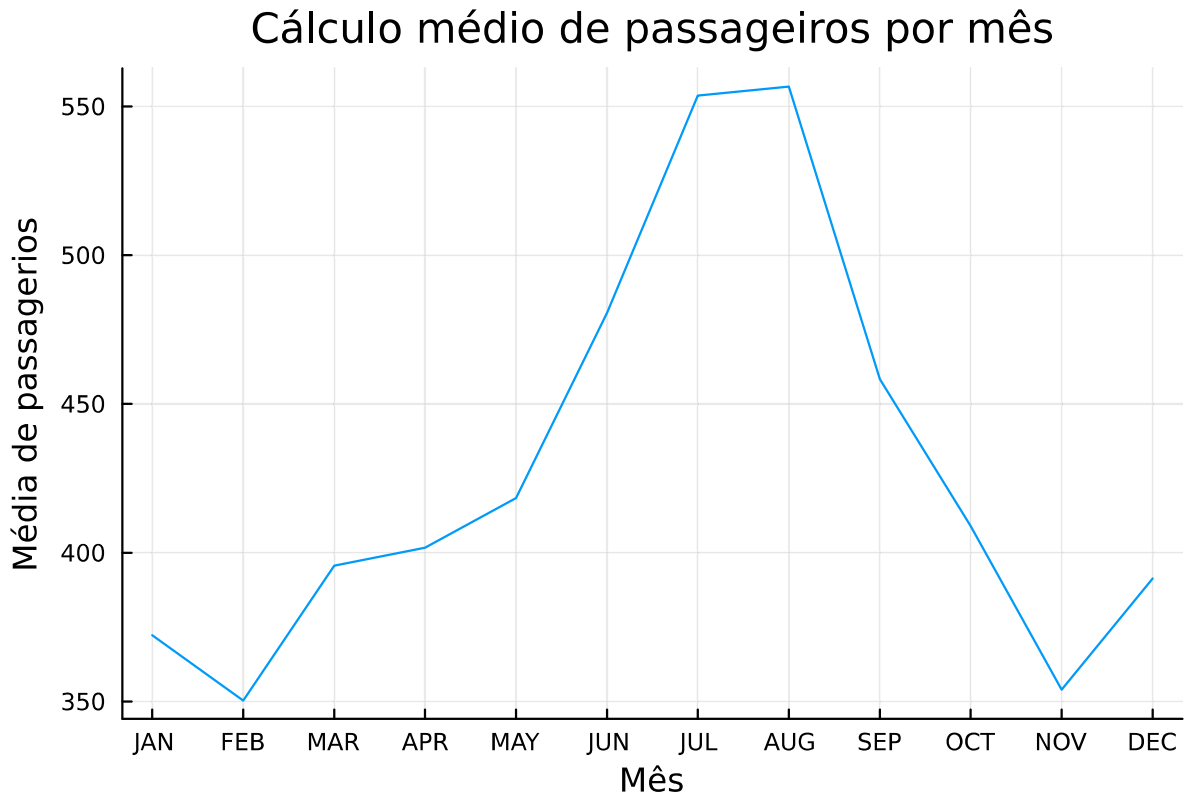
```
df.MeanPassengers = mean_passengers[:]
```

```
mean_passengers_vec =
  [372.333, 350.333, 395.667, 401.667, 418.333, 480.667, 553.667, 556.667, 458.333, 409.0, 3
```

```
mean_passengers_vec = vec(mean_passengers)
```

```
[372.333, 350.333, 395.667, 401.667, 418.333, 480.667, 553.667, 556.667, 458.333, 409.0, 3
```

```
df.MeanPassengers = mean_passengers_vec
```



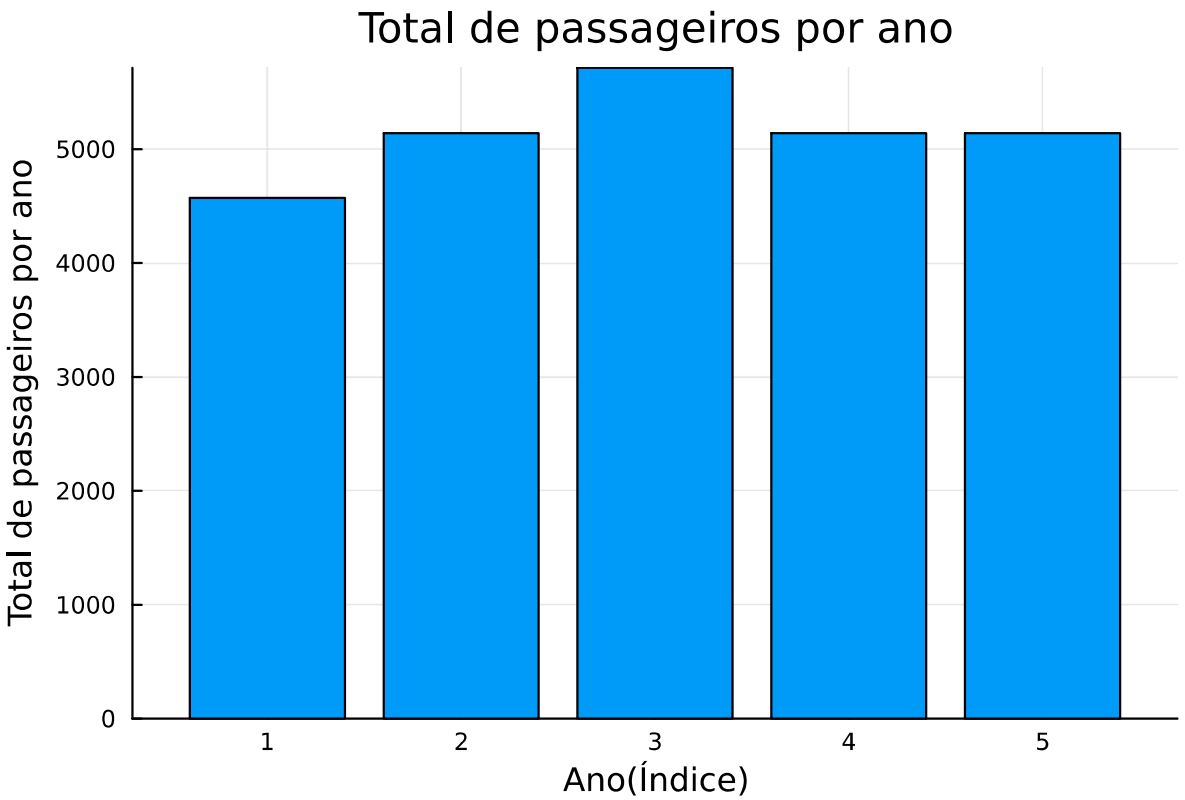

```
plot(df.Month, df.MeanPassengers, title ="Cálculo médio de passageiros por mês",
xlabel="Mês", ylabel="Média de passagerios", legend=false)
```

```
total_passengers_per_year =  [4572.0, 5140.0, 5714.0, 5142.0, 5142.0]
```

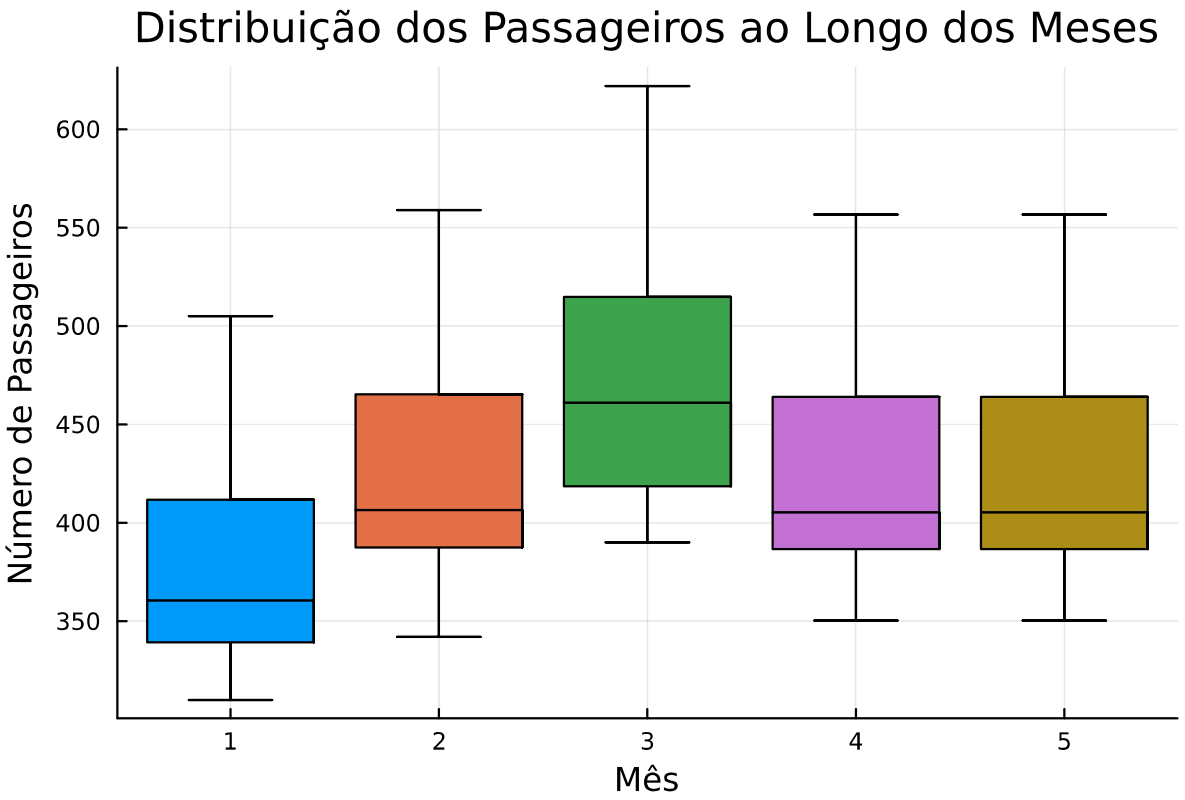
```
total_passengers_per_year = sum(Matrix(df[:, 2:end]), dims=1)[:]
```

```
["Month", "1958", "1959", "1960", "MeanPassengers_vec", "MeanPassengers"]
```

```
names(df)
```



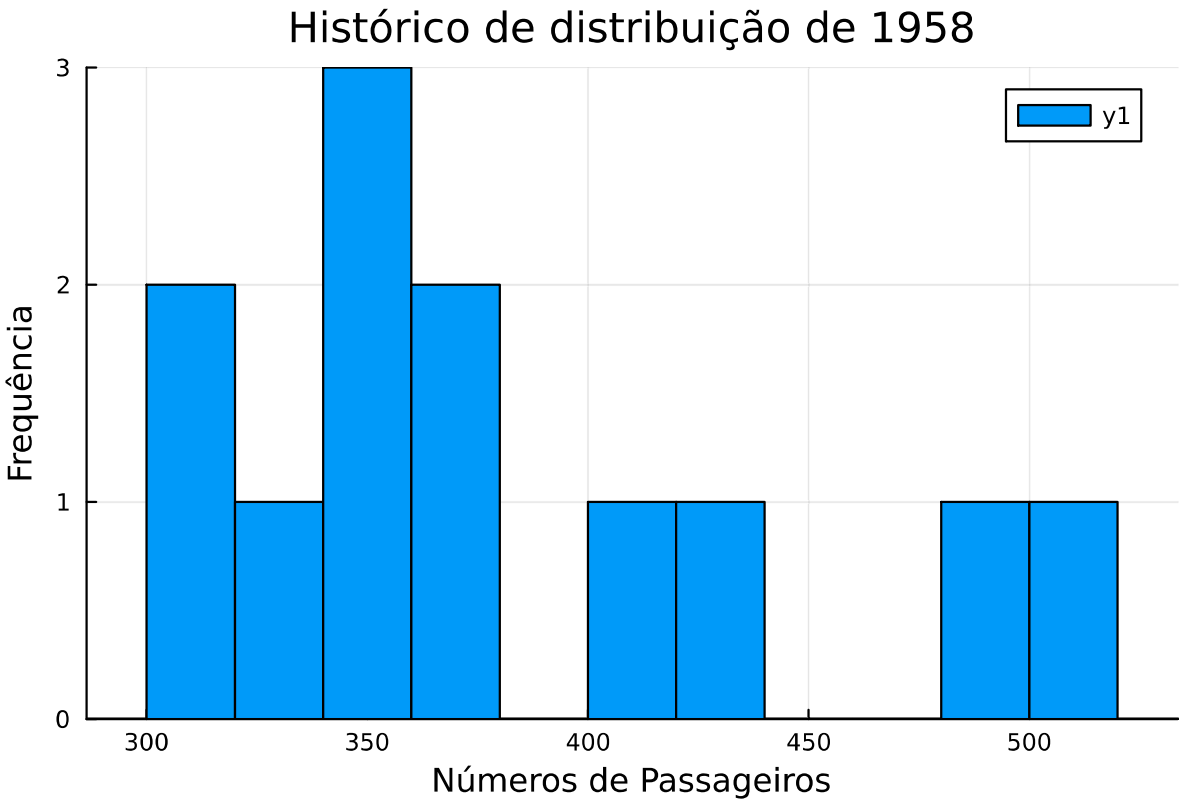

```
bar(1:length(total_passengers_per_year), total_passengers_per_year,
    title="Total de passageiros por ano",
    xlabel="Ano(Índice)",
    ylabel="Total de passageiros por ano",
    legend=false)
```




```
boxplot(Matrix(df[:, 2:end]),
        title="Distribuição dos Passageiros ao Longo dos Meses",
        xlabel="Mês",
        ylabel="Número de Passageiros",
        legend=false)
```

```
selected_year_data =  [340, 318, 362, 348, 363, 435, 491, 505, 404, 359, 310, 337]
```

```
selected_year_data = df[!, "1958"]
```




```
histogram(selected_year_data, bins=10,
        title="Histórico de distribuição de 1958",
        xlabel="Números de Passageiros", ylabel="Frequência")
```